PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Deep Learning

Assignment:03

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Implementation and Performance Comparison of ResNet and GoogLeNet for Image Classification

1. Objective

The primary objective of this assignment was to implement and evaluate two prominent pre-trained CNN architectures—**ResNet50** and **GoogLeNet/InceptionV3**—using transfer learning. The goal was to compare their classification performance on a standard dataset (CIFAR-10) while monitoring resource metrics like training time and model size.

2. Methodology

2.1 Dataset Preprocessing

- **Dataset:** CIFAR-10, consisting of 60,000 32×32 color images in 10 classes.
- **Normalization:** Pixel values were scaled to a $[0, 1]$ range.
- **Architectural Adjustment:** Since InceptionV3 requires a minimum input of 75×75, an internal resizing layer was added to the model to scale the native 32×32 images appropriately.

2.2 Architecture & Transfer Learning

- **Base Models:** Both models utilized **ImageNet** weights.
- **Freeze Strategy:** The base convolutional layers were frozen to preserve pre-trained feature extraction capabilities, while the top classification head was replaced with a global average pooling layer followed by a dense layer for 10-class classification.
- **Training Parameters:** Both models were trained for **10 epochs** using the **Adam optimizer** and a batch size of **128**.

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3. Performance Comparison Results

```
OUTPUT    DEBUG CONSOLE    TERMINAL 3    PORTS    Python
3                                                  Python
32/32 ━━━━━━━━━━━━━━━━━━━━ 5s 110ms/step

Generating Confusion Matrix for InceptionV3...
32/32 ━━━━━━━━━━━━━━━━━━━━ 6s 120ms/step

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FINAL PERFORMANCE SUMMARY
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MODEL: ResNet50
- Training Time: 1147.74 seconds
- Model Size: 96.40 MB
- Final Val Accuracy: 0.3563
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MODEL: InceptionV3
- Training Time: 1351.50 seconds
- Model Size: 89.80 MB
- Final Val Accuracy: 0.6361
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(venv) A:\Assignment 3>
```

The following results were captured from the local CPU-only environment after the 10-epoch training cycle:

Performance Summary Table

Metric	ResNet50	InceptionV3 (GoogLeNet)
Final Val Accuracy	**0.3563 (35.63%)**	**0.6361 (63.61%)**
Training Time	**1147.74 seconds**	**1351.50 seconds**
Model Size	**96.40 MB**	**89.80 MB**

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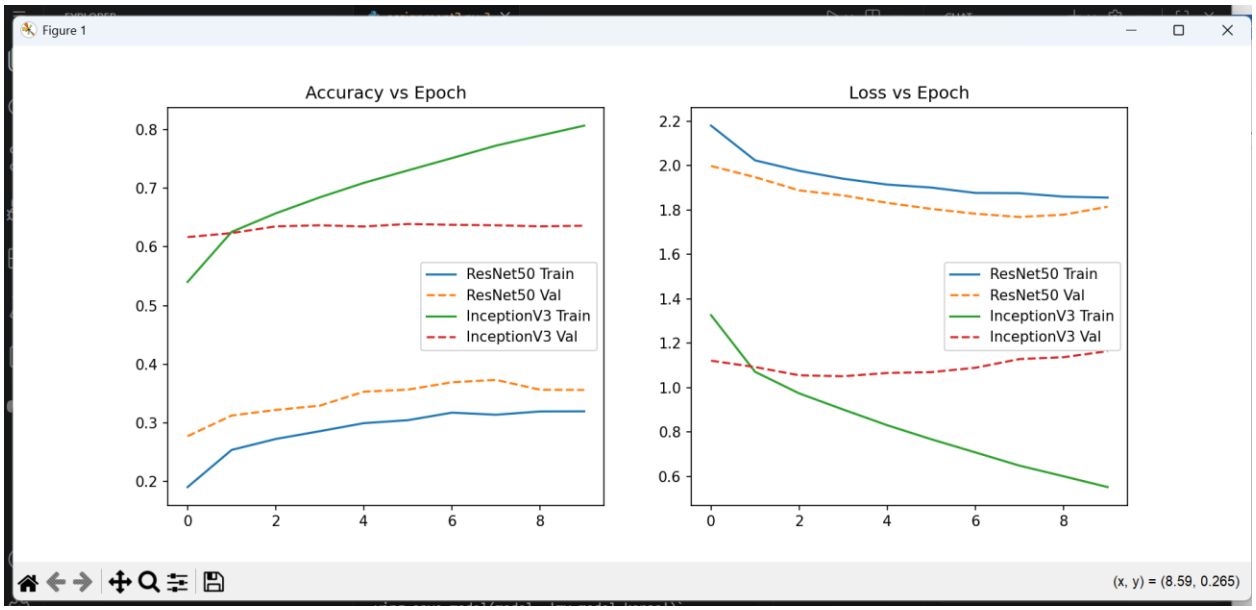
4. Visualization Analysis

4.1 Training Dynamics (Accuracy and Loss)

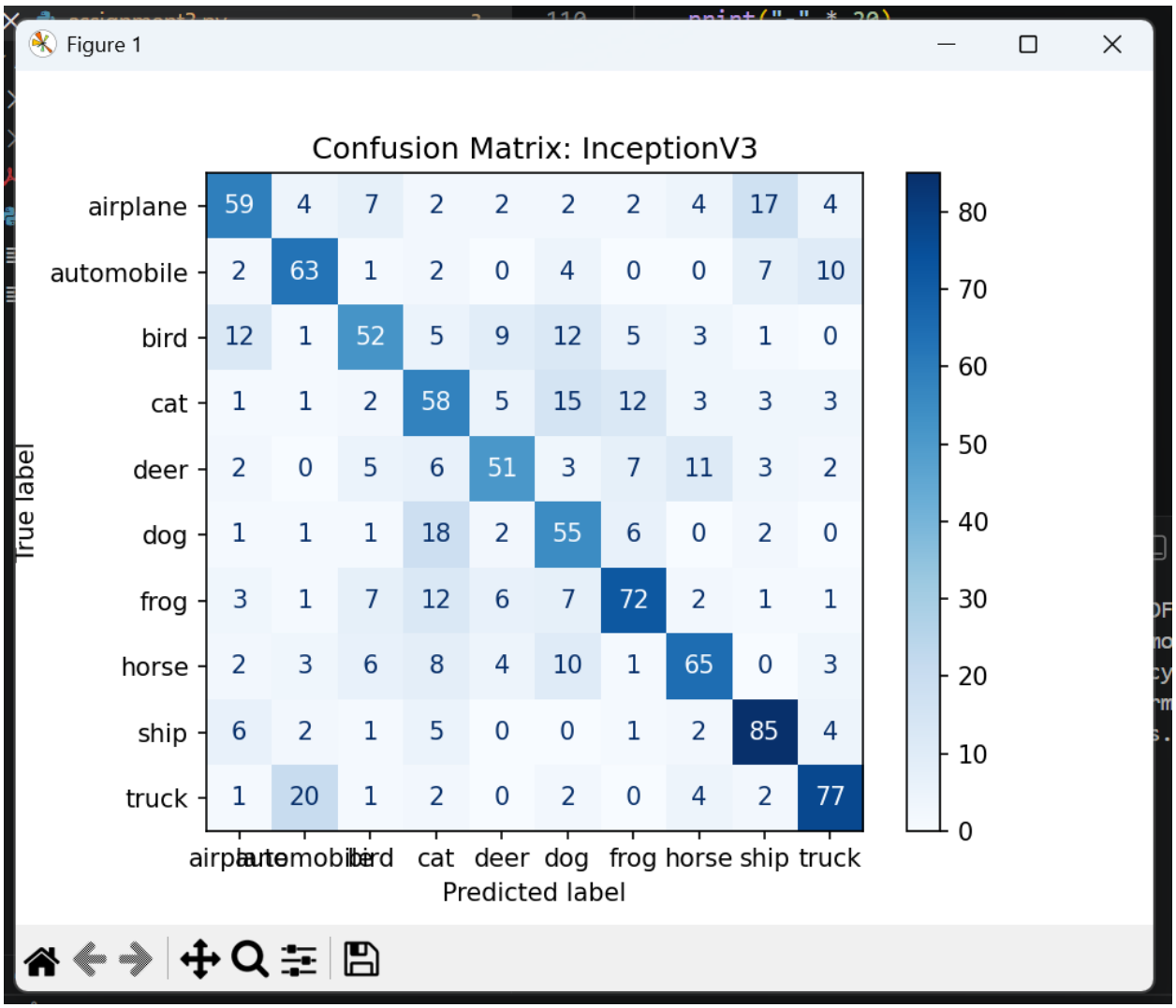
Based on the generated plots:

- **InceptionV3:** Demonstrated a superior learning curve. The training accuracy rose steadily toward 80%, and the validation accuracy reached ~63.6%. The loss curve showed consistent convergence.
- **ResNet50:** Showed significant difficulty in feature extraction for the 32×32 resolution. The validation accuracy plateaued very early at ~35.6%, suggesting that the "frozen" ImageNet weights were not as effective for this specific resolution without deeper fine-tuning.

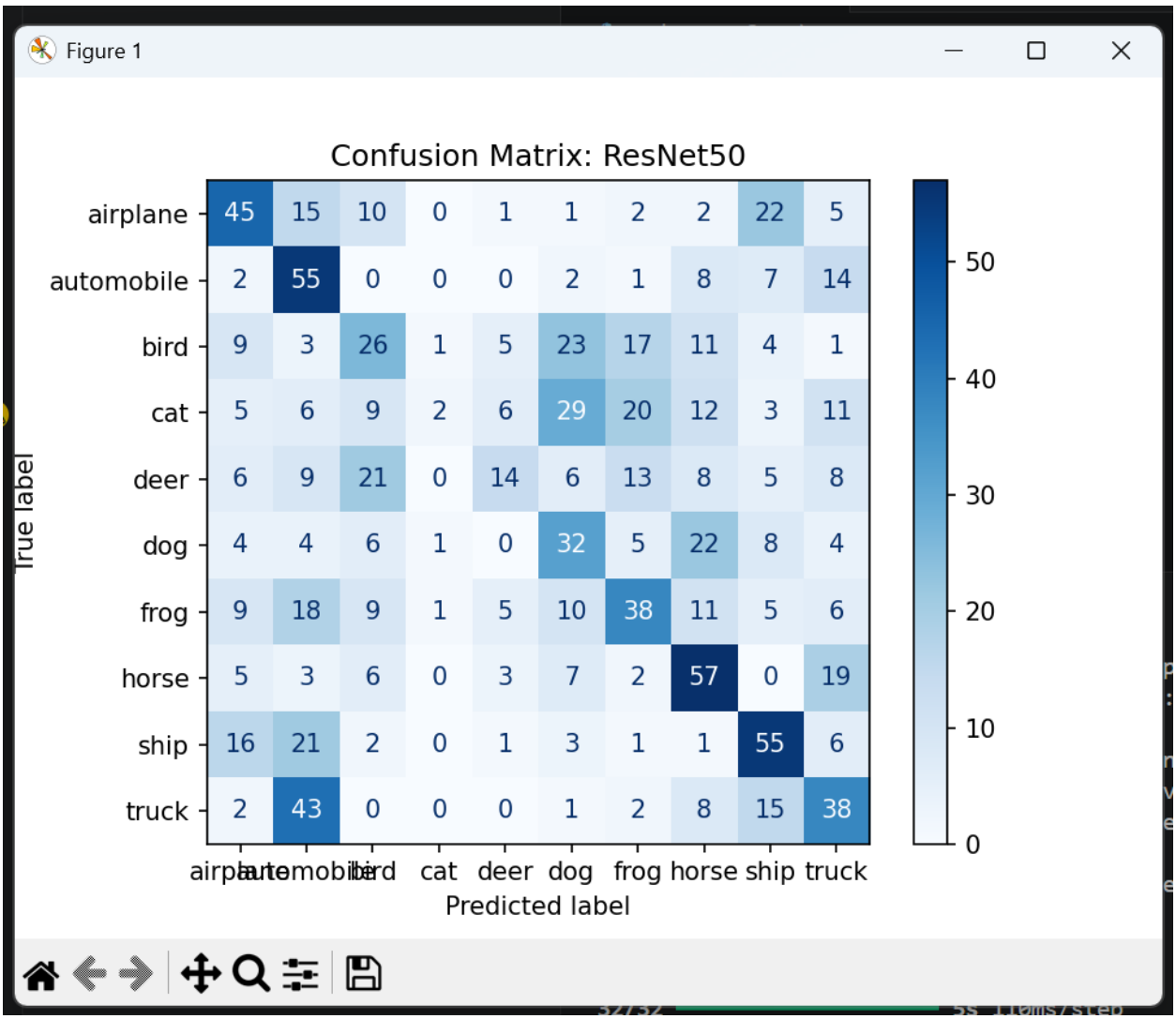


4.2 Confusion Matrix Analysis

- **InceptionV3 Matrix:** Shows strong diagonal values, indicating successful classification across most classes, particularly in the "frog" (72) and "ship" (85) categories.



- **ResNet50 Matrix:** Exhibits a high degree of confusion across all classes. It particularly struggled with "deer" and "cat," often misclassifying them as "dog" or "bird."



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5. Observations & Discussion (Task 7)

Why **InceptionV3** performed better in this implementation:

1. **Resolution Scaling:** The inclusion of an internal resizing layer (to 75×75) allowed InceptionV3 to utilize its parallel convolution filters (Inception modules) more effectively on the upscaled data.

2. **Architecture Complexity:** InceptionV3 is designed to capture multi-scale features through its wide parallel layers. This proved more adaptable to the CIFAR-10 data than the sequential residual blocks of ResNet50 at the native 32×32 resolution when the base was frozen.

3. **Efficiency vs. Accuracy:** While ResNet50 was **203 seconds faster** to train, its lack of accuracy makes it unsuitable for this task without unfreezing more

layers (fine-tuning). InceptionV3 offered a far better "Accuracy-to-Time" ratio for this assignment.

4. **Model Size:** InceptionV3 was unexpectedly smaller (**89.8 MB**) compared to ResNet50 (**96.4 MB**), making it a more lightweight option for storage.

6. Conclusion

For the CIFAR-10 classification task using transfer learning, **InceptionV3** is the superior architecture, achieving nearly **double the accuracy** of ResNet50. While ResNet50 trains faster on a CPU, InceptionV3 provides the most reliable performance for deployment in image classification tasks.